



INTERNSHIELD

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CLOUD COMPUTING & SECURITY

LAB 01

PERFORMING S3 BUCKET ENUMERATION

Discover and identify exposed S3 buckets across the internet using industry standard tools.



RECONNAISSANCE

Identify publicly accessible S3 buckets



ENUMERATION

Discover bucket names and configurations



ANALYSIS

Assess exposure and permissions



SECURITY INSIGHT

Understand risks of misconfigured storage



CLOUD SECURITY

Strengthen your cloud security mindset

PRACTICAL LABORATORY MANUAL

HANDS-ON CLOUD SECURITY EXERCISES



AUTHORIZED TRAINING ENVIRONMENT

This lab is designed for educational purposes only. Perform all exercises only in authorized lab environments and with explicit permission.



LAB INFORMATION

FIELD	DETAIL
LAB	1
TITLE	Perform S3 Bucket Enumeration Using Various S3 Bucket Enumeration Tools
PRIMARY PLATFORM	Amazon Web Services
SERVICE	Amazon S3
PRIMARY TOOL	Lazys3
LAB TYPE	Cloud Security Assessment
FOCUS	S3 Bucket Enumeration

LAB SCENARIO

Enumeration tools are used to collect detailed information about target systems in order to assess and, where misused, exploit them. Information gathered by S3 enumeration tools typically consists of a list of misconfigured S3 buckets that are available publicly. Attackers who discover such buckets can gain unauthorized access to them, and may modify, delete, or exfiltrate the stored content.

In this lab, you will use enumeration tools to obtain as much information as possible about the target cloud environment. This lab demonstrates S3 bucket enumeration tools that help extract a list of publicly available S3 buckets so that exposure can be identified and remediated.

AUTHORIZED LAB ENVIRONMENT

This exercise must be performed only against the cloud resources and targets explicitly provided or authorized for the training environment.

Never perform S3 enumeration against third-party organizations, production systems, or public targets without explicit authorization.

LAB OBJECTIVES

- Enumerate S3 Buckets using Lazys3

The learner will understand how S3 enumeration tools identify potentially exposed bucket names, and why publicly accessible storage represents a cloud-security risk.

TOOL OVERVIEW

Lazys3

Lazys3 is a Ruby script tool used to brute-force AWS S3 bucket names using different permutations. It can identify publicly accessible S3 buckets and can also be used to search for bucket names associated with a specified company name.

SECURITY RELEVANCE

- Bucket-name discovery
- Public exposure identification
- Cloud-storage assessment
- Security posture analysis

LAB ENVIRONMENT & PREREQUISITES

Requirement	Purpose
Parrot Security	Security-testing workstation
MATE Terminal	Command-line interface
Ruby	Required runtime for Lazys3
Lazys3	S3 enumeration tool
Authorized Cloud Target	Controlled assessment target

SECURITY & AUTHORIZATION WARNING

S3 bucket enumeration can reveal cloud-storage resources that belong to other organizations.

Perform this laboratory ONLY against:

- Instructor-provided targets
- Dedicated training infrastructure
- Resources explicitly authorized for testing

Do not enumerate third-party organizations or production resources.

TASK 1

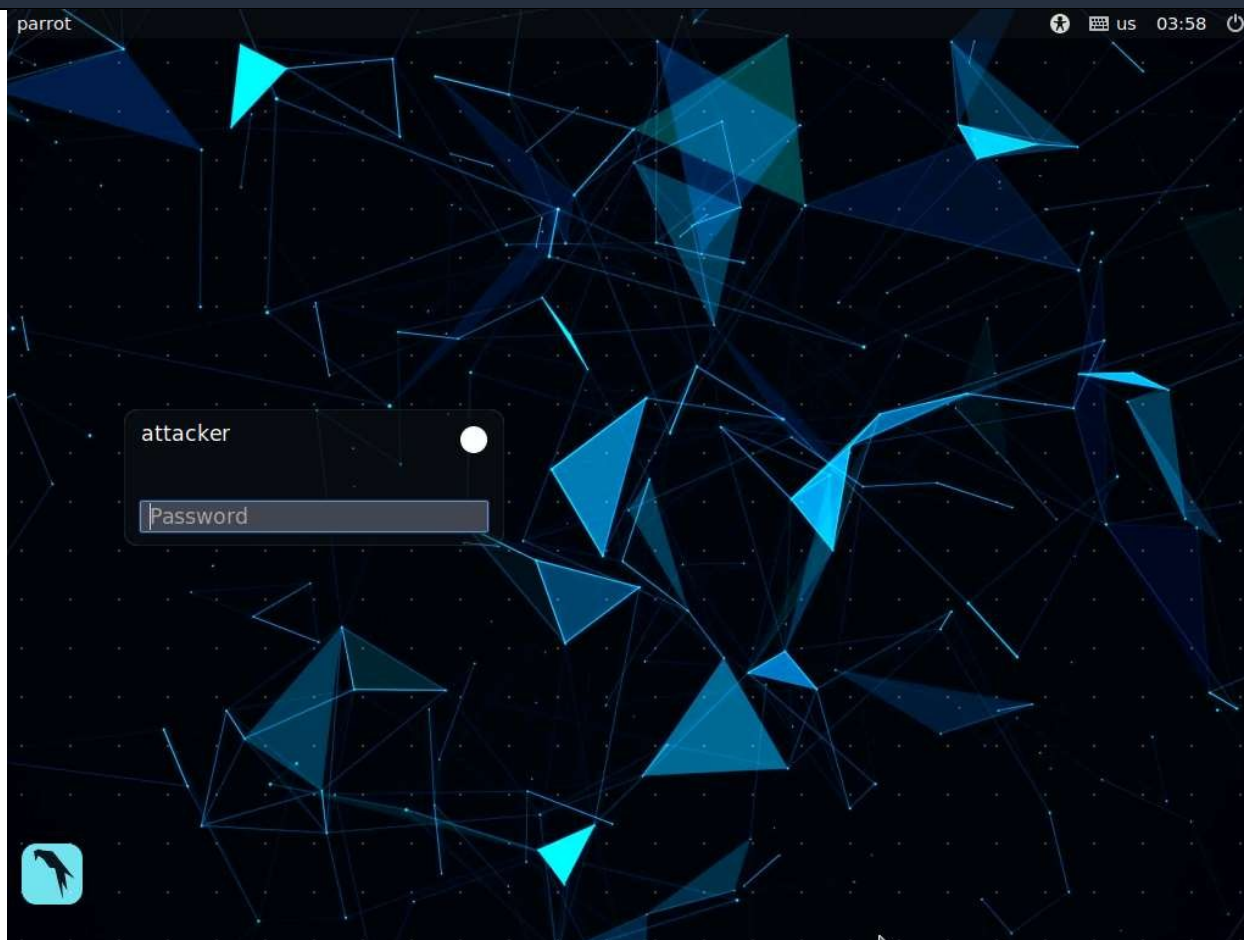
ENUMERATE S3 BUCKETS USING LAZYS3

OBJECTIVE

Identify publicly accessible S3 bucket names using Lazys3 within the authorized training environment, and document the resulting information for security assessment.

PRACTICAL PROCEDURE

STEP 01 ACCESS THE PARROT SECURITY MACHINE



ACTION

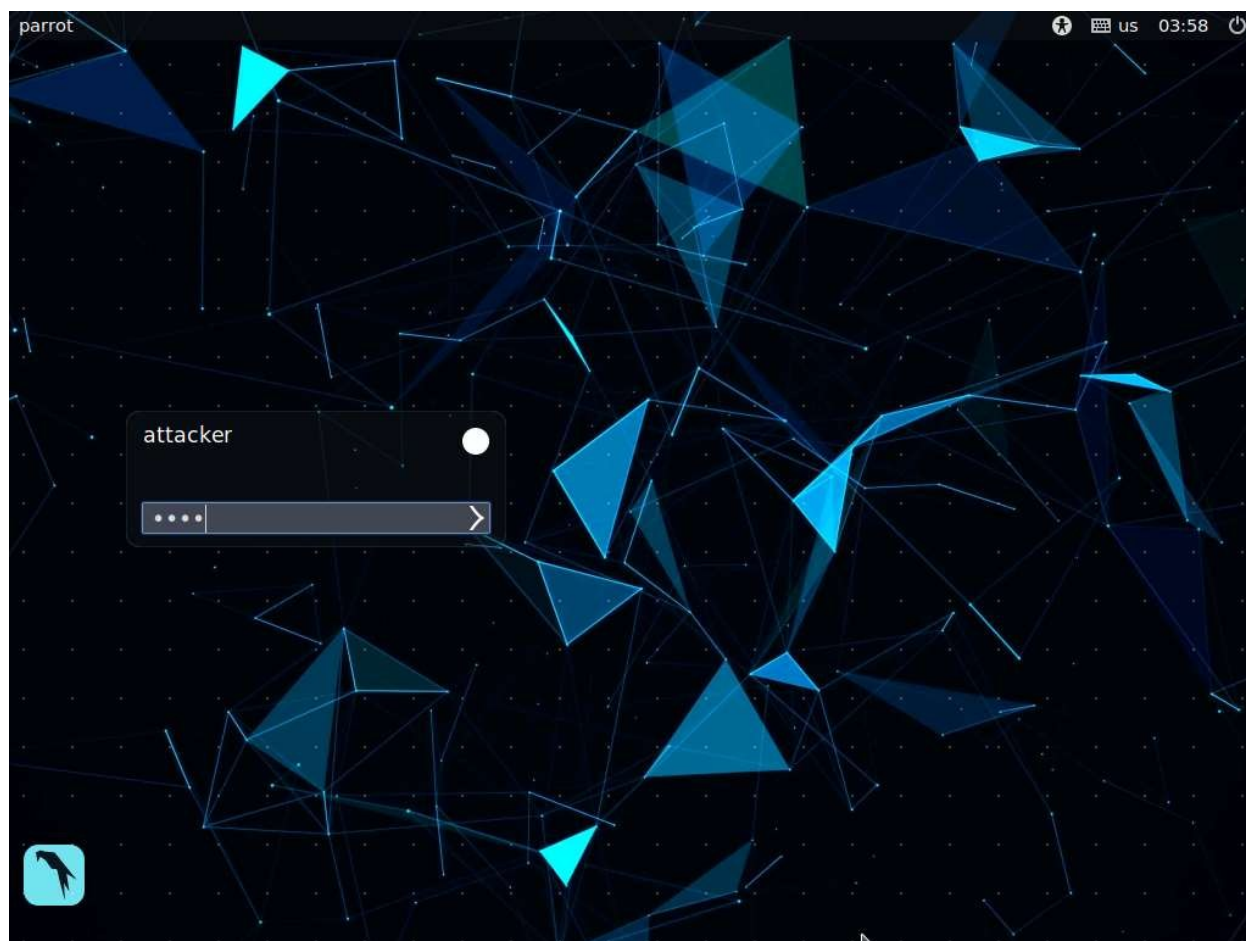
Click Parrot Security to switch to the Parrot Security machine.

EXPECTED RESULT

The Parrot Security machine becomes the active workstation.

SECURITY INSIGHT

Working from a dedicated security workstation keeps the assessment environment isolated from production systems.



STEP 02 LOG IN TO PARROT SECURITY

ACTION

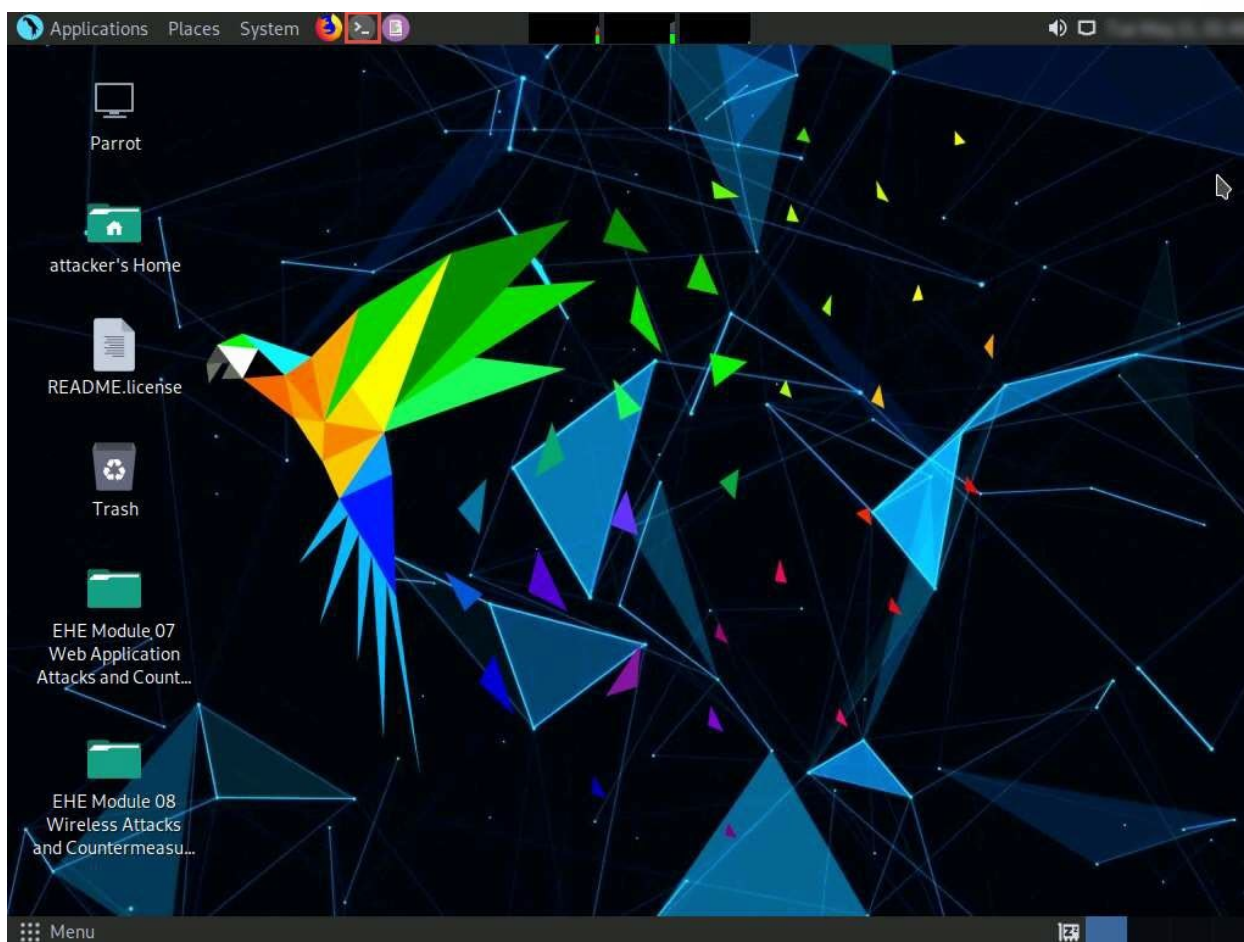
In the login page, the attacker username will be selected by default. Enter the password toor in the Password field and press Enter to log in to the machine.

EXPECTED RESULT

Parrot Security starts successfully and the desktop loads.

SECURITY INSIGHT

Authenticating with the designated lab credentials keeps the session within the authorized training scope.

**STEP 03 OPEN A TERMINAL WINDOW****ACTION**

Click the MATE Terminal icon at the top of the Desktop window to open a Terminal window.

EXPECTED RESULT

A Terminal window opens on the desktop.

SECURITY INSIGHT

The terminal is the primary interface for running enumeration tools such as Lazys3.

STEP 04 SWITCH TO THE ROOT USER**ACTION**

LAB ONE | S3 BUCKET ENUMERATION

In the Terminal window, type `sudo su` and press Enter to run the programs as a root user. If a pop-up window appears asking to update the machine, click No to close it.

```
sudo su
```

EXPECTED RESULT

The shell prompt changes to indicate root-level access.

SECURITY INSIGHT

Some enumeration tools require elevated privileges to install dependencies or bind resources; elevate only within the authorized lab session.

STEP 05 PROVIDE THE SUDO PASSWORD

ACTION

In the [sudo] password for attacker field, type `toor` as the password and press Enter. The password will not be visible as you type.

EXPECTED RESULT

Root access is granted in the terminal.

SECURITY INSIGHT

Password masking is a standard terminal security behavior that prevents shoulder-surfing of credentials.

STEP 06 NAVIGATE TO THE ROOT DIRECTORY

ACTION

Type `cd` and press Enter to jump to the root directory.

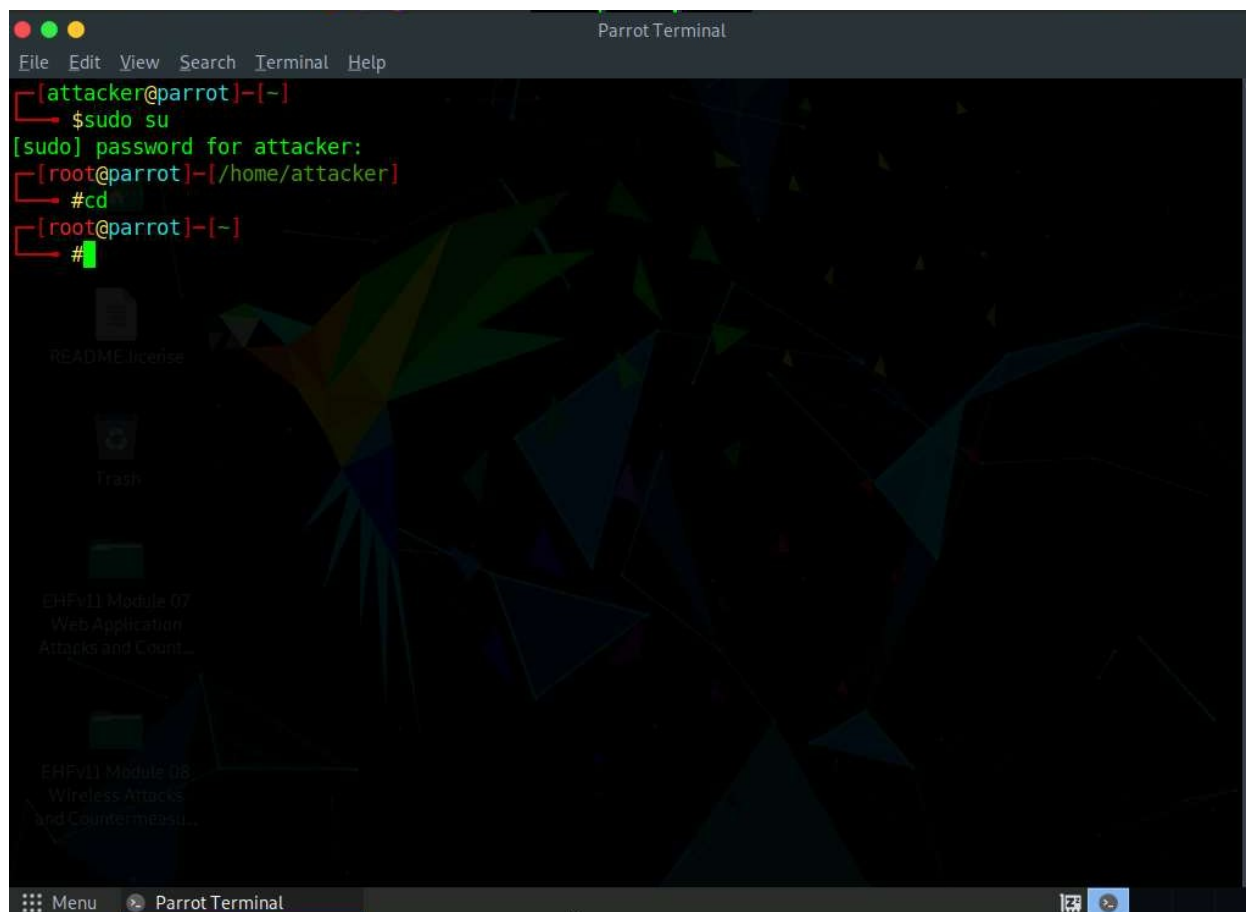
```
cd
```

EXPECTED RESULT

The terminal prompt now reflects the root directory location.

SECURITY INSIGHT

Confirming your working directory before running tools helps avoid path-related execution errors.

**STEP 07 NAVIGATE TO THE LAZYS3 DIRECTORY****ACTION**

Type `cd lazys3` and press Enter to navigate to the cloned repository. The Lazys3 tool has already been downloaded in the lab setup.

```
cd lazys3
```

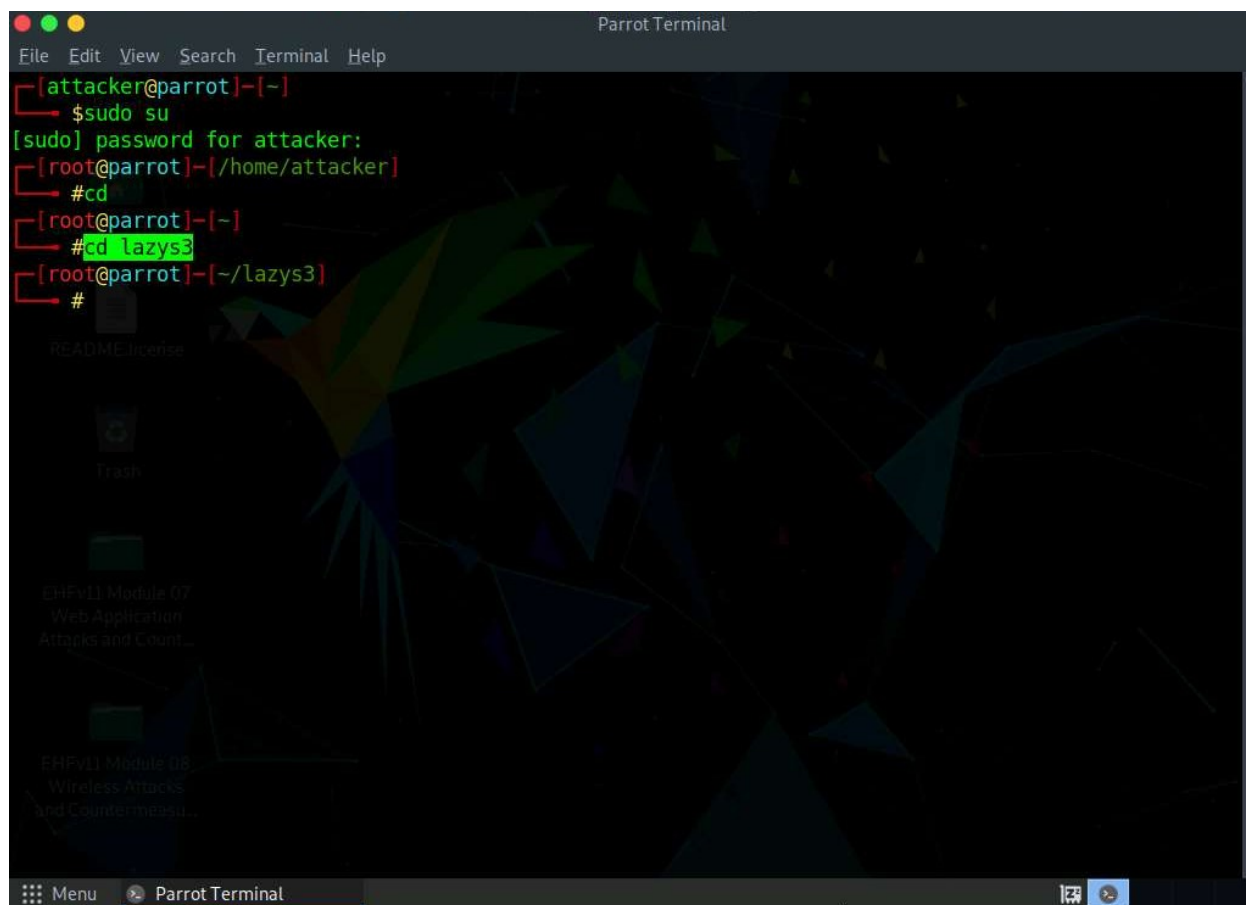
EXPECTED RESULT

The prompt changes to the lazys3 directory.

SECURITY INSIGHT

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Enumeration tools are typically distributed as source repositories; navigating to the correct directory ensures the script and its dependencies are found.



```
Parrot Terminal
File Edit View Search Terminal Help
[attacker@parrot]~$ sudo su
[sudo] password for attacker:
[root@parrot]~/home/attacker$ #cd
[root@parrot]~$ #cd lazys3
[root@parrot]~/lazys3$ #
```

STEP 08 LIST THE DIRECTORY CONTENT**ACTION**

In the lazys3 folder, type ls and press Enter to list the folder content.

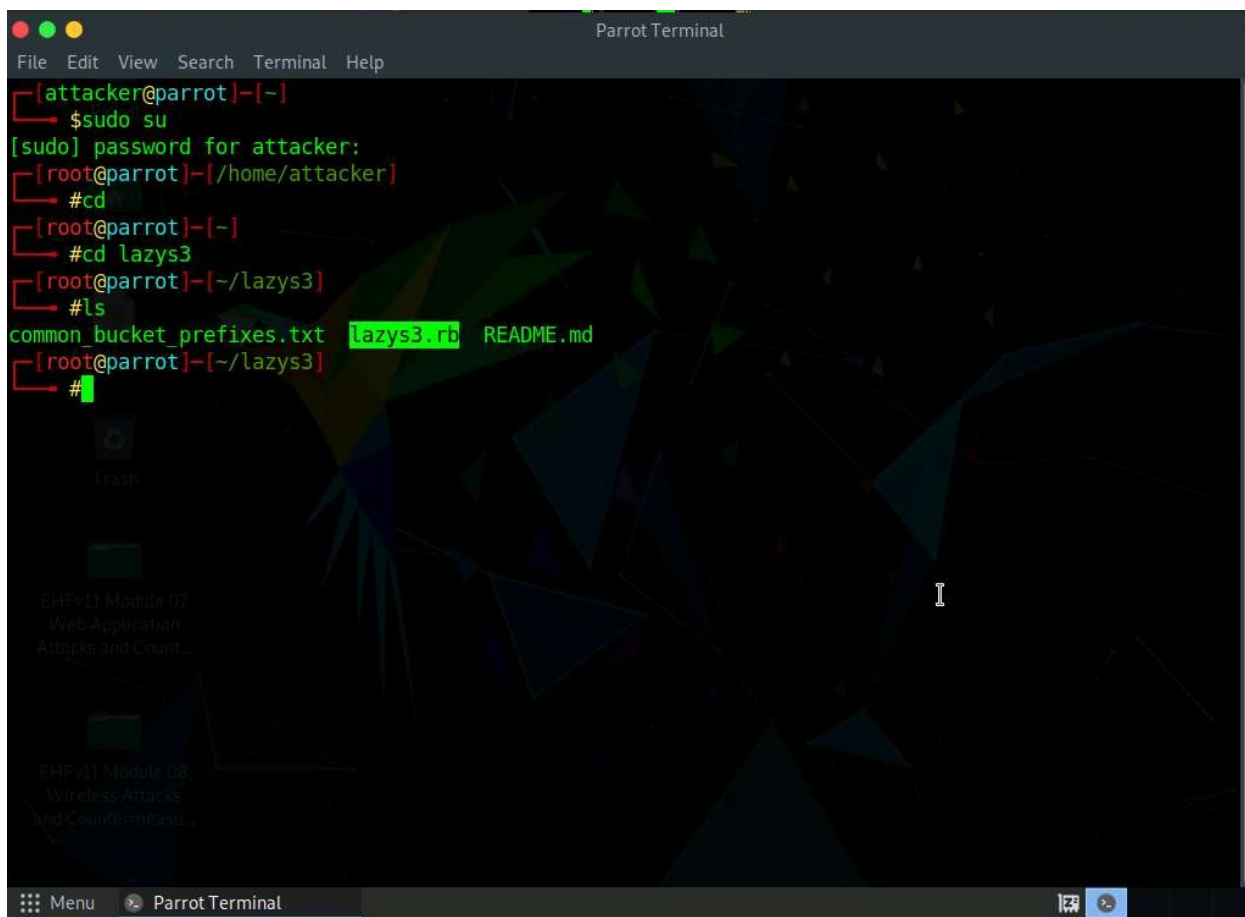
```
ls
```

EXPECTED RESULT

The folder content is displayed.

SECURITY INSIGHT

Reviewing directory contents before execution confirms the required script and supporting files are present.



```
Parrot Terminal
File Edit View Search Terminal Help
[attacker@parrot]~$ sudo su
[sudo] password for attacker:
[root@parrot]~/home/attacker$ #cd
[root@parrot]~$ #cd lazys3
[root@parrot]~/lazys3$ #ls
common_bucket_prefixes.txt lazys3.rb README.md
[root@parrot]~/lazys3$ #
```

STEP 09 CONFIRM THE LAZYS3 SCRIPT IS PRESENT**ACTION**

Review the displayed folder content. The Lazys3 Ruby script is available for the S3 bucket enumeration activity.

EXPECTED RESULT

The Lazys3 files are visible and ready to execute.

SECURITY INSIGHT

Verifying tool integrity before use is a good assessment practice.

STEP 10 RUN THE LAZYS3 SCRIPT**ACTION**

Run the Lazys3 script according to the authorized lab workflow: type `ruby lazys3.rb` and press Enter.

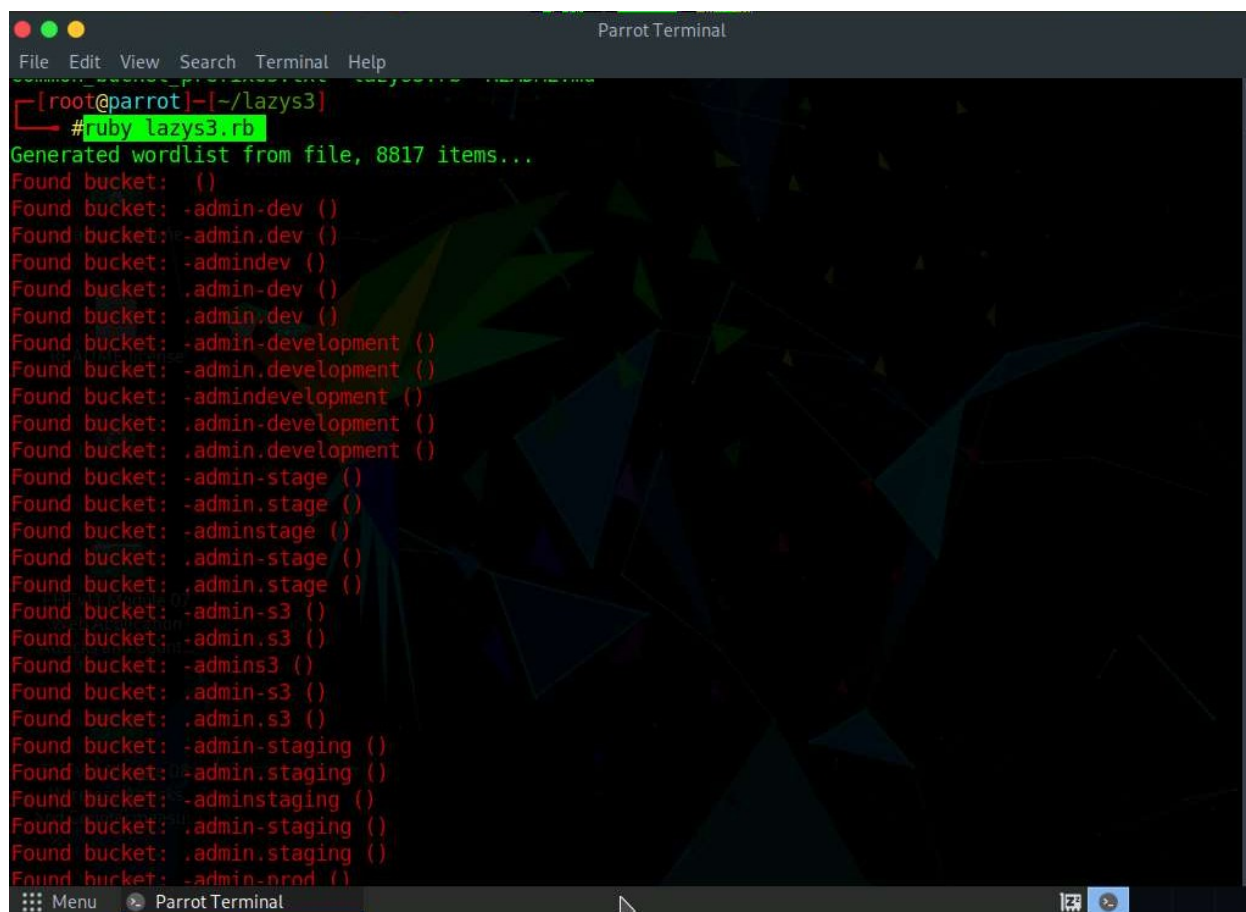
```
ruby lazys3.rb
```

EXPECTED RESULT

Lazys3 executes in the authorized lab environment and begins enumeration.

SECURITY INSIGHT

Running the base script performs a permutation-based brute-force search for common publicly exposed bucket names.



```
ParrotTerminal
File Edit View Search Terminal Help
[root@parrot]--[~/lazys3]
#ruby lazys3.rb
Generated wordlist from file, 8817 items...
Found bucket: ()
Found bucket: -admin-dev ()
Found bucket: -admin.dev ()
Found bucket: -admindev ()
Found bucket: .admin-dev ()
Found bucket: .admin.dev ()
Found bucket: -admin-development ()
Found bucket: -admin.development ()
Found bucket: -admindevelopment ()
Found bucket: .admin-development ()
Found bucket: .admin.development ()
Found bucket: -admin-stage ()
Found bucket: -admin.stage ()
Found bucket: -adminstage ()
Found bucket: .admin-stage ()
Found bucket: .admin.stage ()
Found bucket: -admin-s3 ()
Found bucket: -admin.s3 ()
Found bucket: -admin.s3 ()
Found bucket: -admin.s3 ()
Found bucket: -admin-staging ()
Found bucket: -admin.staging ()
Found bucket: -adminstaging ()
Found bucket: .admin-staging ()
Found bucket: .admin.staging ()
Found bucket: -admin-prod ()
```

STEP 11 REVIEW THE ENUMERATION OUTPUT**ACTION**

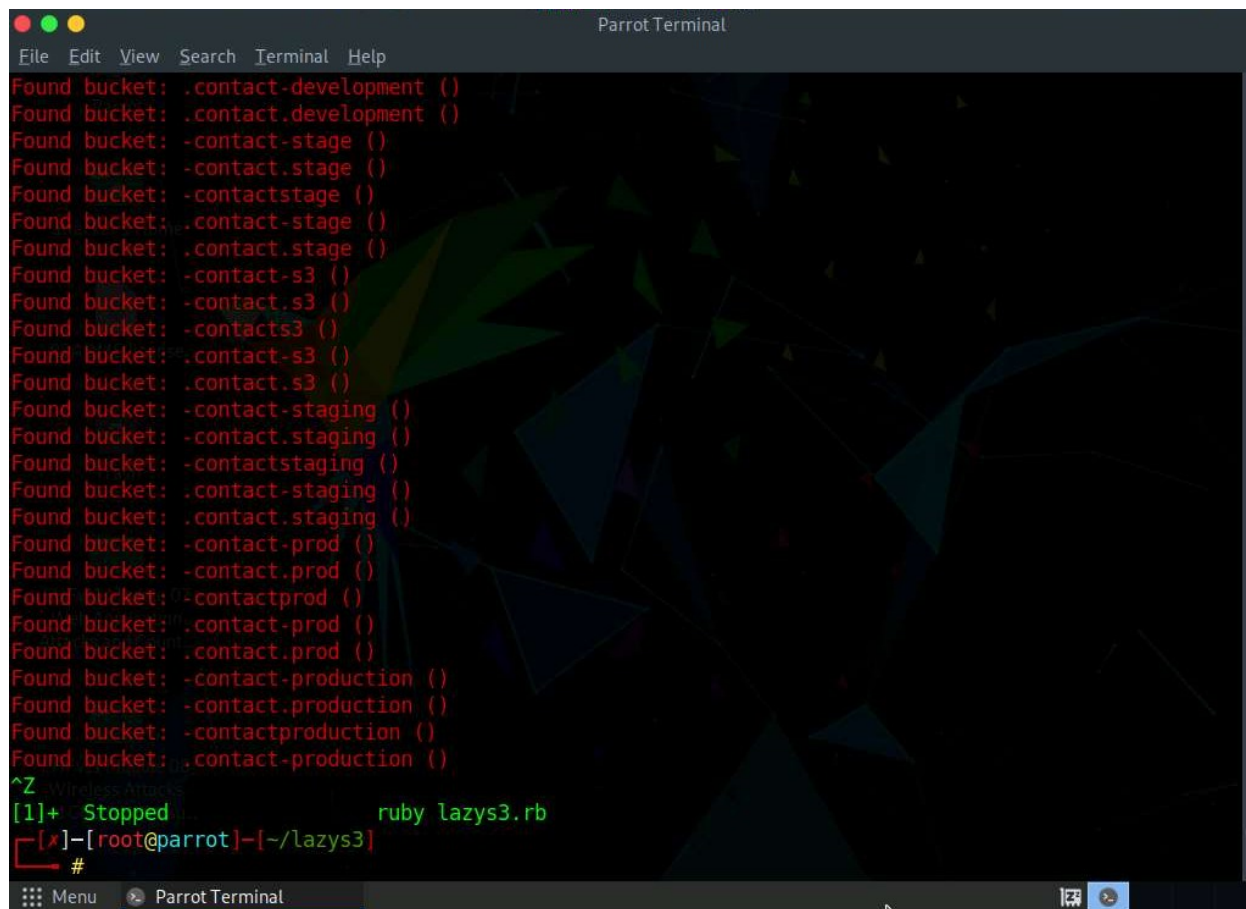
A list of publicly accessible S3 buckets is displayed. Record only the information permitted by the laboratory scope.

EXPECTED RESULT

Publicly accessible bucket information is displayed and can be reviewed.

SECURITY INSIGHT

Bucket names alone can reveal organizational or infrastructure details, so results should be handled and documented carefully.



```
Parrot Terminal
File Edit View Search Terminal Help
Found bucket: .contact-development ()
Found bucket: .contact-development ()
Found bucket: -contact-stage ()
Found bucket: -contact.stage ()
Found bucket: -contactstage ()
Found bucket: .contact-stage ()
Found bucket: -contact-s3 ()
Found bucket: -contact.s3 ()
Found bucket: -contacts3 ()
Found bucket: .contact-s3 ()
Found bucket: -contact-staging ()
Found bucket: -contact.staging ()
Found bucket: -contactstaging ()
Found bucket: .contact-staging ()
Found bucket: -contact-prod ()
Found bucket: -contact.prod ()
Found bucket: -contactprod ()
Found bucket: .contact-prod ()
Found bucket: -contact-production ()
Found bucket: -contact.production ()
Found bucket: -contactproduction ()
Found bucket: .contact-production ()
^Z Wireless Attacks
[1]+  Stopped ruby lazys3.rb
[x]-[root@parrot]-[~/lazys3]
#
```

STEP 12 STOP THE ENUMERATION PROCESS

ACTION

Stop the enumeration process using the terminal control specified by the lab environment: press Ctrl+Z.

Ctrl+Z

EXPECTED RESULT

The running enumeration process is stopped.

SECURITY INSIGHT



Enumeration should run only long enough to gather the information needed for the assessment scope.

STEP 13 SEARCH BUCKETS FOR AN AUTHORIZED TRAINING NAME

ACTION

Search for S3 bucket names associated with a specified company name. For an authorized training organization or instructor-provided test name, use: `ruby lazys3.rb [Company]`. Do NOT use real third-party organizations unless explicit authorization has been provided.

```
ruby lazys3.rb [Company]
```

EXPECTED RESULT

Lazys3 executes a targeted search against the specified training-scope name.

SECURITY INSIGHT

Targeted searches demonstrate how attackers might pivot from a known company name to its cloud footprint, underscoring the importance of naming-convention hygiene.

STEP 14 REVIEW THE TARGETED SEARCH RESULTS

ACTION

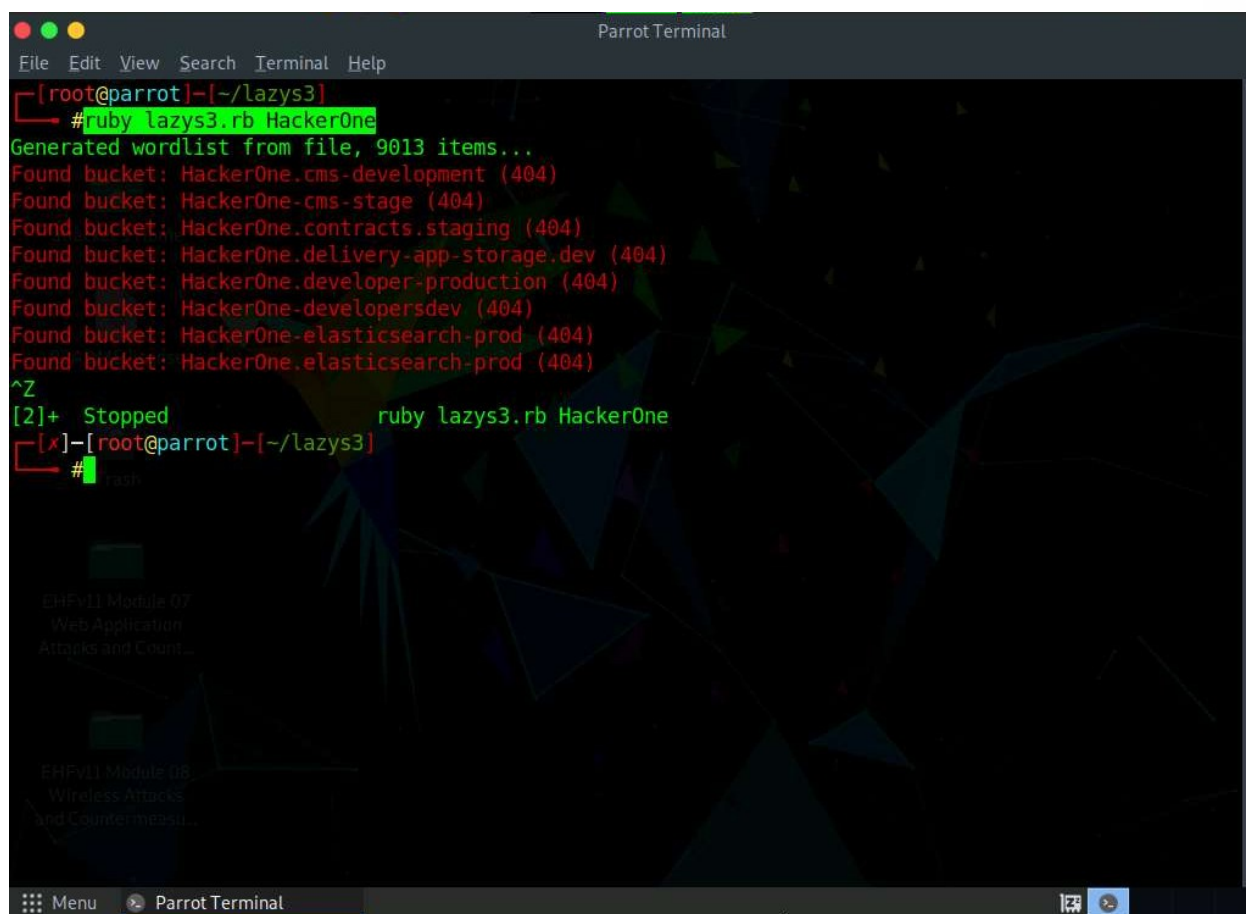
The result appears, showing the obtained list of S3 buckets associated with the specified training name. The number and names of buckets may vary depending on the authorized laboratory environment.

EXPECTED RESULT

Training-scope bucket results can be reviewed.

SECURITY INSIGHT

Result variability across environments illustrates why enumeration findings must always be documented per-assessment rather than assumed from prior runs.



```
Parrot Terminal
File Edit View Search Terminal Help
[root@parrot]--[~/lazys3]
# ruby lazys3.rb HackerOne
Generated wordlist from file, 9013 items...
Found bucket: HackerOne.cms-development (404)
Found bucket: HackerOne.cms-stage (404)
Found bucket: HackerOne.contracts.staging (404)
Found bucket: HackerOne.delivery-app-storage.dev (404)
Found bucket: HackerOne.developer-production (404)
Found bucket: HackerOne-developersdev (404)
Found bucket: HackerOne-elasticsearch-prod (404)
Found bucket: HackerOne-elasticsearch-prod (404)
^Z
[2]+ Stopped ruby lazys3.rb HackerOne
[x]--[root@parrot]--[~/lazys3]
#
```

STEP 15 STOP THE TARGETED ENUMERATION

ACTION

Stop the running enumeration process using the terminal control specified by the lab environment: press Ctrl+Z.

Ctrl+Z

EXPECTED RESULT

The enumeration process can be stopped.

SECURITY INSIGHT

Cleanly terminating tool execution avoids leaving unnecessary background processes running.

STEP 16 CONCLUDE THE ENUMERATION DEMONSTRATION

LAB ONE | S3 BUCKET ENUMERATION

ACTION

This concludes the demonstration of enumerating publicly accessible S3 buckets.

EXPECTED RESULT

The enumeration demonstration is complete.

SECURITY INSIGHT

Formally closing out the technical activity helps transition cleanly into documentation and analysis.

STEP 17 CLOSE WINDOWS AND DOCUMENT FINDINGS

ACTION

Close all open windows and document the acquired information.

EXPECTED RESULT

Findings can be documented and the session is closed.

SECURITY INSIGHT

Thorough documentation of enumeration findings is essential for follow-up remediation and reporting.

SECURITY ANALYSIS

Publicly accessible S3 buckets can expose cloud-stored data and may create confidentiality, integrity, or availability risks for the organizations that own them.

- Public exposure — buckets reachable without authentication
- Misconfiguration — overly permissive bucket policies or ACLs
- Information disclosure — sensitive files, backups, or credentials left accessible
- Data access risk — unauthorized parties viewing stored objects
- Data modification risk — unauthorized changes to stored objects
- Data deletion risk — unauthorized removal of stored objects
- Data exfiltration risk — bulk download of exposed content

S3 SECURITY COUNTERMEASURES

- Apply least privilege to all S3 access.
- Avoid unnecessary public access on buckets.
- Review bucket policies regularly.
- Review ACL configuration for unintended exposure.
- Use IAM controls to scope access precisely.
- Enable appropriate logging and monitoring.
- Review storage permissions on a recurring schedule.
- Use dedicated training buckets for security exercises.
- Remove unused resources promptly.
- Protect sensitive data with encryption and access controls.

S3 BUCKET	ACCESS CONTROL	IAM / POLICY	MONITORING	SECURE STORAGE
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LEARNING OUTCOMES

After completing this lab, students should be able to:

- 1. Explain S3 bucket enumeration.
- 2. Explain why public S3 buckets create security risk.
- 3. Describe the purpose of Lazys3.
- 4. Navigate the Parrot Security environment.
- 5. Execute the supplied lab workflow in an authorized environment.
- 6. Interpret enumeration output at a high level.
- 7. Document cloud-security findings.
- 8. Recommend S3 security controls.

LAB OBSERVATION

Activity	Observation	Security Significance
Parrot Security Login		
Terminal Access		
Lazys3 Directory		
Lazys3 Execution		
Bucket Enumeration		
Training Organization Search		
Enumeration Results		
Final Documentation		

LAB COMPLETION CHECKLIST

- ☐ Parrot Security started
- ☐ Authorized scope confirmed
- ☐ Terminal opened
- ☐ Lazys3 directory accessed
- ☐ Lazys3 files reviewed
- ☐ Enumeration activity completed
- ☐ Training-scope results documented
- ☐ Sensitive information checked / redacted
- ☐ Enumeration process stopped
- ☐ Findings documented
- ☐ Security implications analyzed
- ☐ S3 countermeasures reviewed

REVIEW QUESTIONS

- 1. What is S3 bucket enumeration?
- 2. Why can publicly accessible S3 buckets create security risks?
- 3. What is the purpose of Lazys3?
- 4. Why must S3 enumeration be performed only against authorized targets?
- 5. What types of information can an S3 enumeration assessment reveal?
- 6. What is the difference between discovery and exploitation?
- 7. What security controls can reduce accidental S3 exposure?

TROUBLESHOOTING

Problem	Possible Cause	Recommended Action
Parrot Security does not start	Virtual machine resources unavailable or lab environment still loading	Wait for the environment to finish loading, then retry; contact lab support if the issue persists
Terminal does not open	Desktop environment not fully loaded	Wait for the desktop to finish loading before clicking the terminal icon
Lazys3 directory is unavailable	Repository not present in the expected location	Confirm the lab setup completed correctly; contact the lab administrator
Ruby is unavailable	Ruby runtime not installed or not on PATH	Verify the Ruby environment is installed as part of the lab setup
Lazys3 script does not execute	Missing dependencies or incorrect working directory	Confirm you are in the lazys3 directory and required dependencies are present
No results are displayed	Search scope returned no matching public buckets	This is a valid outcome; document the result as part of the assessment
Results differ from the example	Bucket availability varies by environment and over time	This is expected; document the results observed in your own environment
Enumeration takes longer than expected	Network conditions or permutation volume	Allow the process to continue, or stop it using the specified terminal control if scope permits

BEST PRACTICES

- Always define the assessment scope before testing.
- Use dedicated training resources.
- Never enumerate third-party infrastructure without permission.
- Never store discovered sensitive information unnecessarily.
- Redact sensitive information in screenshots.
- Document findings professionally.
- Apply least privilege to S3.
- Review public-access settings regularly.
- Remove temporary training resources after the lab.

LAB SUMMARY

This laboratory introduced the process of identifying publicly accessible S3 bucket information using an enumeration tool. Enumeration is the reconnaissance stage of a broader cloud-security lifecycle: findings from this stage feed directly into assessment, documentation, remediation, and verification activities.

DISCOVER	ASSESS	DOCUMENT	REMEDIATE	VERIFY
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